CHAPTER 1

Syllabus Map & Strategy

GATE 2027 COA syllabus · what to study · priority order · how these notes are organised

QUICK TRICK

The whole GATE 2027 COA syllabus is **six lines**: Instruction set & addressing modes · ALU design · Control unit (hardwired / microprogrammed) · Memory interfacing & hierarchy (performance, cache mapping) · I/O (interrupt, DMA) · Pipelining & hazards.
Roughly **8–10 marks** per paper (typical, not official). Almost all of it is **calculation**: practise numbers.

1.1 The official syllabus (GATE 2027, IIT Madras)

● **LEVEL 1 · BEGINNER**

Section 3 of the CS paper, word for word from the official syllabus PDF (gate2027.iitm.ac.in):

```
Instruction set and addressing modes.
Design of arithmetic and logic unit (ALU).
Design of control unit - hardwired and microprogrammed.
Memory interfacing and hierarchy: performance, cache memory mapping.
I/O interface (interrupt and DMA).
Instruction pipelining, pipeline hazards.
```

Also in the official file, but under **Section 2 (Digital Logic)**: "Number representation and arithmetic (fixed and floating point)". COA questions use it constantly, so Chapter 3 covers it here. If you already have it in your Digital Logic notes, use Chapter 3 as revision.

Always confirm the final wording on the official site before the exam. Earlier years used a differently worded COA list (for example "memory hierarchy: cache, main memory and secondary storage"), so disk topics are not named now. They are kept as a short Chapter 11 because they are cheap marks if asked.

1.2 Where the marks come from

● **LEVEL 2 · EXAM**

This tier list is my own estimate from the usual pattern of GATE papers, not an official statement. Use it only to order your time.

Tier	Topics	Notes chapter
Tier 1 (very high)	Pipelining (speedup, stalls, hazards) · cache mapping and cache performance · addressing modes and instruction formats	8 · 10 · 4, 5
Tier 2 (high)	IEEE 754 and number representation · CPU performance (CPI, Amdahl) · memory performance and chip interfacing · DMA and interrupt calculations · microprogrammed control word size	3 · 2 · 9 · 12 · 7
Tier 3	ALU (Booth, adders) · disk access time · hardwired control	6 · 11 · 7

GATE POINT

Most 2-mark COA questions are **numerical answer type (NAT)** or multi-step cache/pipeline problems. There is no negative marking for NAT and MSQ, so always attempt.

1.3 Chapter map of these notes

● LEVEL 1 · BEGINNER

Ch	Title	Syllabus clause covered
2	CPU Performance	Performance (CPI, MIPS, Amdahl)
3	Number Representation & Arithmetic	Digital Logic section: fixed and floating point
4	Instruction Set & Formats	Instruction set
5	Addressing Modes	Addressing modes
6	ALU Design	Design of ALU
7	Control Unit: Hardwired & Microprogrammed	Design of control unit
8	Instruction Pipelining & Hazards	Pipelining, hazards
9	Memory Hierarchy, Performance & Interfacing	Memory interfacing and hierarchy: performance
10	Cache Memory Mapping	Cache memory mapping
11	Disk & Secondary Storage	Memory hierarchy (older wording)
12	I/O: Interrupt & DMA	I/O interface (interrupt and DMA)
13	Master Formula Sheet & Mixed Practice	Revision

1.4 Study order and strategy

● LEVEL 2 · EXAM

1. CPU performance (Ch 2) → 2. Pipelining (Ch 8) → 3. Cache (Ch 9, 10) → 4. ISA and addressing modes (Ch 4, 5) → 5. Number representation (Ch 3) → 6. I/O (Ch 12) → 7. Control unit and ALU (Ch 7, 6) → 8. Disk (Ch 11).

COMMON MISTAKE

- Treating COA as theory. Practise with units (bits vs bytes, KB = 2^{10}, ns vs cycles).
- Mixing "words" and "bytes" when counting address bits.
- Skipping the number-representation chapter because it sits in Digital Logic.

1.5 Books

Hamacher (Computer Organization) · Patterson & Hennessy (Computer Organization and Design; Computer Architecture) · Morris Mano (Computer System Architecture). Solve previous-year GATE questions after each chapter.

Next: Chapter 2 — CPU Performance (CPI, MIPS, Amdahl's law).